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Institutional Remedies for Individual Expert Identification

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Why Social Epistemology?

- ▶ In a modern society, everyone knows a lot:
 - ▶ We live on a round planet that circles the sun in an enormous almost empty space called the universe.
 - ▶ The planet has been warming up rapidly in the past century and CO₂, among other greenhouse gases, is causing it.
 - ▶ Human body is made up of trillions of cells and almost every cell has the same DNA.
- ▶ Only very few people have direct evidence for the claims above. Most people learn through others: textbooks, schoolteachers, documentary films, newspapers, etc.
- ▶ This applies to most knowledge held by most people in the modern society.

Varieties of Social Epistemology

Classification by Alvin Goldman (2011)

- ▶ Individual Doxastic Agents with Social Evidence
 - ▶ Expert Identification (Goldman 2001, 2016)
- ▶ Collective Doxastic Agents
 - ▶ Group Belief (Gilbert 1987, 1989, 2004, Bird 2014, 2022)
- ▶ System-Oriented Social Epistemology
 - ▶ Credit Economy in Science (Kitcher 1990, Strevens 2003, Zollman 2018)
 - ▶ My research lies here.

Epistemic Systems: What and Why

- ▶ An epistemic system (or institution) is a system (or institution) that serves epistemic purpose.
- ▶ Why?
- ▶ Epistemic tasks of importance are often done, not just by a team or a group, but by a well institutionalized system.
- ▶ Better understanding of the impact of different institutional arrangement can help design better systems.

Epistemic Systems: Some Examples

- ▶ Ship Navigation: *Cognition in the Wild* (1995)
- ▶ Science
 - ▶ Done by the Scientific Community
 - ▶ Scientists are trained for years to follow the protocols and to be part of the community
- ▶ Medicine
- ▶ Finance
- ▶ Voting

Epistemic Identification

The Problem

When facing a purported experts, how do you make the decision whether to trust her or not?

When experts offer competing opinions, which one should you trust?

- ▶ Difficulty: Novice cannot comprehend experts' evidence and argument due to a lack of professional training.
- ▶ Necessity of Epistemic Deference: It's often rational to suspend one's own judgment and defer to experts. (Hardwig, 1985, 1991)
- ▶ Goldman (2001): should use past track records of the expert's cognitive success, which is retrospectively verifiable (or "exoteric") to the novice.

Epistemic Systems

The Paper

- ▶ I use three game-theoretic models to demonstrate that verifiable track records can be misleading as well.
 - ▶ Purported experts can strategically build up their track records by tackling easier cognitive challenges.
 - ▶ This strategic movement's impacts cannot be avoided by the novice by himself, but can be mitigated by institutional arrangements.
- ▶ This shows that the problem of expert identification is not just a problem of individual doxastic agent with social evidence, but also a problem of institution-oriented social epistemology.

Toy Model: Setup

Adapted from Kamenica and Gentzkow (2011)

- ▶ Agents: Expert, Novice (both Bayesian)
- ▶ State of Interest: Expert's competency $\omega \in \{0, 1\}$, where 1 means competent, 0 means incompetent.
- ▶ Timeline:
 - ▶ Expert and Novice starts with common prior P_0
 - ▶ Expert decides whether and how to reveal some true information to the Novice.
 - ▶ Novice forms posterior P_I .
 - ▶ Novice decides whether to trust Expert based on P_I .
- ▶ Initial Information
 - ▶ Common Knowledge: 30% of all purported experts are competent.
 - ▶ (For simplification) Expert has no private information about his competency.
 - ▶ Thus, common prior $P_0(\omega = 1) = 0.3$.

Toy Model: Agents

- ▶ Common Prior: $P_0(\omega = 1) = 0.3$
- ▶ Novice Posterior: P_I
- ▶ Novice Decision/Action (common knowledge): $a = A(P_I, t)$
 - ▶ Novice makes a decision based on his posterior P_I and a threshold t . Let $t = 2/3$ for our example.
 - ▶ $a = 1$ (i.e. to trust) if $P_I(\omega = 1) \geq t$; $a = 0$ otherwise.
- ▶ Novice Objective: The Novice wants to make the right decision, i.e. to maximize the expectation of his decision accuracy: $EU_{Novice} = 1 - E[|\omega - a|]$.
- ▶ Expert Objective: Expert wants to be trusted, regardless of her true competency. That is, to maximize her chance of being trusted $EU_{Expert} = E[a]$.

Toy Model: Learning

Bayesian Updating / Conditionalization

► Bayes Rule: $P(H|E) = \frac{P(E|H) \cdot P(H)}{P(E)}$

- $P(\cdot)$ is prior distribution.
- $P(\cdot|E)$ is posterior distribution.
- $P(\cdot|H)$ is likelihood function.

► Law of Total Probability:

$$P(E) = P(E|H)P(H) + P(E|\neg H)P(\neg H)$$

► Together: $P(H|E) = \frac{P(E|H) \cdot P(H)}{P(E) = P(E|H)P(H) + P(E|\neg H)P(\neg H)}$

Toy Model: Learning

The Martingale Property / Reflection Principle

- ▶ Martingale property (Reflection Principle): The expectation of posterior equals prior.
- ▶ $E[P(H|E) = P(H)]$ for any H, E .
- ▶ Derivation: Let $H \in \{H_1, H_2\}$ and $E \in \{E_1, E_2\}$.

$$\begin{aligned} E[P(H_1|E)] &= P(E_1)P(H_1|E = E_1) + P(E_2)P(H_1|E = E_2) \\ &= P(H_1) \end{aligned}$$

Toy Model: No Information Revealed

- ▶ Without further information ($P_I = P_0$), the Novice will not trust the Expert. Thus, $EU_{Expert} = 0$, $EU_{Novice} = 0.7$.
- ▶ The situation can be improved by verifiable/true information revealed by tests.

Toy Model: Test

- ▶ A test reveals true information (potentially with some error).
- ▶ The expert decides whether to take a test and which test to take, but cannot influence the result of the test. I.e., the information revealed is true information.
- ▶ Suppose there only exists one infallible test that reveals the expert's true competency without error. I.e. the test signals 'Pass' if $\omega = 1$ and 'Fail' if otherwise.

	$Pr(Pass \omega)$	$Pr(Fail \omega)$
$\omega = 0$	0	1
$\omega = 1$	1	0

- ▶ Will the expert take the test?
 - ▶ Yes. EU_{Expert} improves from 0 to 0.3
- ▶ It also helps the novice: EU_{Novice} improves from 0.7 to 1

Toy Model: Test Optimal for the Expert

- ▶ Let's be more realistic: tests fail sometimes.
- ▶ A test has two types of errors: false positive (Type 1) and false negative (Type 2). Denote the error rates by ϵ_1 and ϵ_2 respectively. We can represent a test in general as follows.

	$Pr(Pass \omega)$	$Pr(Fail \omega)$
$\omega = 0$	ϵ_1	$1 - \epsilon_1$
$\omega = 1$	$1 - \epsilon_2$	ϵ_2

- ▶ The infallible test is thus ($\epsilon_1 = 0, \epsilon_2 = 0$).
- ▶ We allow the Expert to choose which test to take by specifying ϵ_1 and ϵ_2 .
- ▶ It turns out that ($\epsilon_1 = \frac{3}{14}, \epsilon_2 = 0$) is optimal for the Expert, where his chance of being trusted $EU_{Expert} = 0.45$.
- ▶ The Novice's decision accuracy also improves from the base case, to 0.85, but below the previous case where complete information is revealed.

Toy Model: How So?

- ▶ This test ($\epsilon_1 = 0, \epsilon_2 = 0$) is tailored based on the novice's reaction function to maximize the probability of seeing *Pass* under the constraint that the Novice will trust the expert after seeing *Pass*
- ▶ With $\epsilon_1 = \frac{3}{14}, \epsilon_2 = 0$
 - ▶ $P_I(\omega|Pass) = \frac{2}{3}$ which is exactly at the threshold $t = \frac{2}{3}$.
 - ▶ $P_I(\omega|Fail) = 0$.
- ▶ By controlling ϵ_1 and ϵ_2 , the Expert makes P_I as concentrate on $\frac{2}{3}$ as possible.

Toy Model: How So?

- ▶ Mathematically speaking, the martingale property (reflection principle) only requires $E(P_0) = E(P_I)$ but doesn't further constrain the distribution of P_I .
- ▶ By revealing true information in certain tailored manner, the Expert influences the distribution of Novice's posterior to maximize the chance of her being trusted.
- ▶ Intuitively, false positive error (ϵ_1) helps the Expert gets trusted while false negative error (ϵ_2) does the opposite.
- ▶ When given the opportunity, the expert would like to maximize ϵ_1 and minimize ϵ_2 . $\epsilon_1 = \frac{3}{14}, \epsilon_2 = 0$ is the result of the optimization under the constraint $P_0 = 0.3$.

Toy Model: So What?

- ▶ The true information revealed by testing never decreases the novice's decision accuracy and often improves it.
- ▶ What test differs for the Expert and the Novice.
 - ▶ $(\epsilon_1 = 0, \epsilon_2 = 0)$ for the Novice
 - ▶ $(\epsilon_1 = \frac{3}{14}, \epsilon_2 = 0)$ is optimal for the Expert.
- ▶ Under the reasonable assumption that Expert picks which test to take, there is nothing the novice can do to avoid the sub-optimality.
- ▶ However, the sub-optimality can be avoided via some institutional arrangement, which would be clearer once we move to a slightly more sophisticated model.

Toy Model: Limitation

- ▶ The assumption that the Expert picks tests by controlling its error term ϵ_1 and ϵ_2 is quite unrealistic.
- ▶ In the second model, we improve on this limitation by using a fixed error term for tests.
- ▶ In the second model, the expert picks the difficulty level of the tests instead.

Second Model

Fixed Error Rate and Variant Threshold

- ▶ We are interested in the expert's competency ω . A test reveals a result $S = \omega + \epsilon$, i.e. the expert's competency with an error rate ϵ . For simplicity, we assume that $\epsilon \sim N(0, 0.5)$. The test result S allows the Novice to update his posterior belief in the usual Bayesian manner. $P(\omega|S)$.
- ▶ Because ϵ is normally distributed, S takes real value in $(-\infty, \infty)$, which is not a good model for any realistic test. We discretize the test result S into two categories: pass (1) and fail (2), to make the test more realistic (and possible for the expert to influence the novice).

Second Model

More on Variant Threshold

- ▶ Given the range of S , the threshold can in principle take any real value, which makes unintuitive for us to think about. An alternative (but equivalent) way to think about the threshold is to consider how it influences the Novice's belief.

$$S_x = \begin{cases} 1 & P(\omega = 1|S) > x \\ 0 & P(\omega = 1|S) \leq x \end{cases}$$

- ▶ Now the Expert influences the Novice by choosing a testing standard $x \in [0, 1]$ to influence the Novice's decision.
- ▶ Note: this threshold is known to the Novice.

Second Model

Solution and Discussion

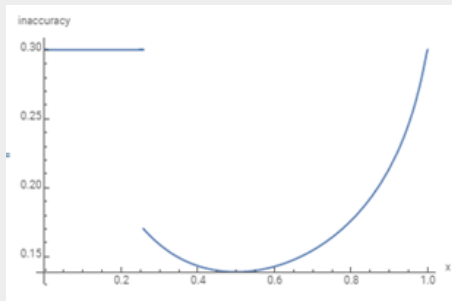
- Solving the optimization problem, we have the following:

	Testing Standard	EU_{Expert}	$1 - EU_{Novice}$
x^*	0.5	0.269435	0.138749
$x^\dagger$	0.257734	0.389462	0.170179

- The strategic phenomena in this model are not different from the previous one.
- But it characterizes tests in a more intuitive way and makes it convenient for us to talk about regulation.

Second Model

Solution and Discussion

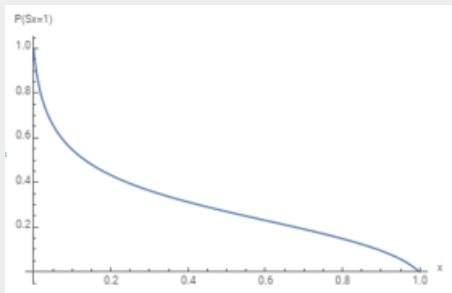


x: Test Difficulty, y: Novice Decision Inaccuracy

- Novice starts to differentiate different test results at $x^\dagger = 0.257734$

Second Model

Solution and Discussion

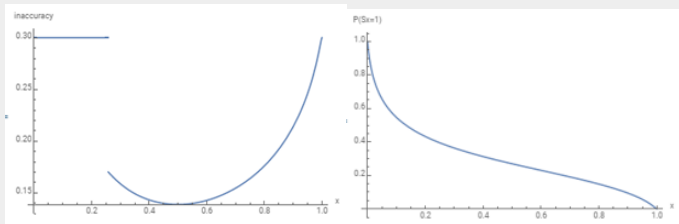


x : Test Difficulty, y : $P(S_x = 1)$

- Novice starts to differentiate different test results at $x^\dagger = 0.257734$

Second Model

Solution and Discussion



x : Test Difficulty; left y : Novice Inaccuracy; right y : $P(\text{Pass})$

- ▶ Minimal Testing Standard:
- ▶ If we set the minimal testing standard at x^* , i.e., forbidding the expert from taking tests $x < x^*$, the expert's optimal choice would be to pick x^* .

Second Model

Limitations

- ▶ Like the Toy model, the Experts strategy is based on tailoring against the Novice's reaction function.
- ▶ This strictly speaking assumes common knowledge about Novice's reaction function, which is unrealistic to some extent.
- ▶ This is not entirely unrealistic though. As long as the Expert has some general imperfect understanding about Novice's reaction function, the strategic phenomena can happen.
- ▶ We introduce further element in the third model, where Expert doesn't need to know anything about the Novice's reaction function.

Third Model: Strategic Disclosure

- ▶ Suppose that I'm a lawyer, I may tell you truthfully that I graduated from Harvard law school without telling you that I was ranked at the bottom in my class.
- ▶ Or I may tell you truthfully that I passed the bar exam but wouldn't tell you that I failed eight times before I did.
- ▶ In the third model, we allow the Expert to choose whether or not to disclose her test results after taking the test.

Third Model

Chance of Failure

- ▶ To make the non-disclosure credible, we need to introduce the genuine possibility of failure in testing, which can be interpreted as the test failed to yield a result.
- ▶ Otherwise, the novice can easily map the Expert's non-disclosure to a bad result. If it's a good result, the expert would have disclose.
- ▶ With this possibility of failure, the novice cannot (at least not automatically) distinguish between a bad result and a null result (where test failed to yield a result).
- ▶ Let the probability of failure in test be 0.02.

Third Model

Chance of Failure

- ▶ With probability 0.02, $S = \emptyset$.
- ▶ Otherwise, $S = \omega + \epsilon$ where $\epsilon \sim N(0, 0.5)$ like before.

$$S_x = \begin{cases} Pass & \text{if } P(\omega = 1|S) > x \\ Fail & \text{if } P(\omega = 1|S) \leq x \\ Null & \text{if } S = \emptyset \end{cases}$$

Third Model

Novice Assumption

- ▶ In this model, how the Novice makes the decision to trust is left unspecified.
- ▶ We instead focus on the accuracy of the Novice's posterior P_I , which clearly is positively correlated with the Novice's decision accuracy.
- ▶ It's possible to discuss P_I because we know that the Novice is a Bayesian agent.
- ▶ So, we assess the epistemic performance of the Novice by the mean squared error (MSE) of his belief, i.e. $E[(\omega - P_I(\omega))^2]$.

Third Model

Decision Structure

- ▶ Expert needs to make two decisions:
 - ▶ Test difficulty x
 - ▶ Disclosure Strategy θ , which maps test results $\{Pass, Fail, Null\}$ to disclosure decisions $\{D, ND\}$.
- ▶ Novice makes one decision:
 - ▶ When a test result is disclosed, the novice updates his posterior in a familiar Bayesian manner $P^D = P(\omega = 1|S_x)$. No decision needed here.
 - ▶ When no test result is disclosed, the novice can no longer update, but must instead pick a posterior P^N .
- ▶ This makes the model technically a game, which we will analyze using Nash equilibrium.

Third Model

Key Observations

- ▶ The key for solution is that this model reduces to something very similar to the second model under Nash Equilibrium.
- ▶ Some basic assumption of Nash Equilibrium, namely common knowledge of player strategies, is crucial for solving this model.
- ▶ Let (x_E, θ_E) be Expert's strategy and P_E^N be Novice's strategy at Nash Equilibrium.
- ▶ By the common knowledge assumption, (x_E, θ_E) is known to the Novice. P^N is thus chosen in a Bayesian manner, namely, $P^N = P(\omega = 1 | \theta_E(S_{x_E}))$.
- ▶ So at equilibrium, the Novice makes a Bayesian update like before.

Third Model

Solution

- Proposition: At Nash equilibrium, Expert chooses x_E, θ_E that minimize the expectation of P^N .

- Proof:

$$(a) E[\theta_E(S_{x_E})P^D(x_E, S_{x_E}) + (1 - \theta_E(S_{x_E}))P_E^N]$$

$$(b) E[E(S_{x_E})P^D(x_E, S_{x_E}) + (1 - \theta_E(S_{x_E}))P(\omega = 1|\theta_E(S_{x_E}) = 0)]$$

$$(c) E[\theta'(S_{x'})P^D(x', S_{x'}) + (1 - \theta'(S_{x'}))P(\omega = 1|\theta'(S_{x'}) = 0)]$$

$$(d) E[\theta'(S_{x'})P^D(x', S_{x'}) + (1 - \theta'(S_{x'}))P_E^N]$$

- (a) = (b), since Novice is Bayesian updating at equilibrium.
- (b) = (c), by Martingale property.
- A deviation is profitable if and only if (d) > (a/b/c).
- if and only if $P_E^N < P(\omega = 1|\theta'(S_{x'}) = 0)$
- Therefore, there is no deviation when P^N is minimized.

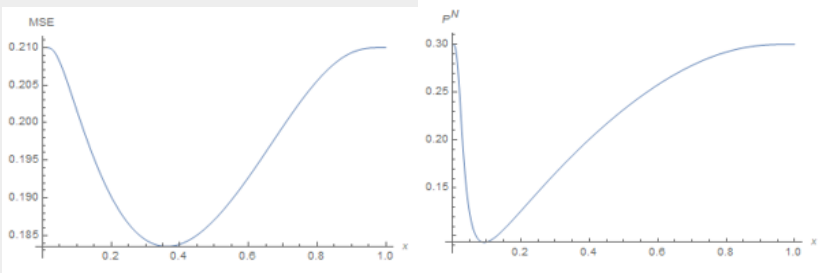
Third Model

Some Key Observations

- ▶ Now we have reduced the problem in to solving for x_E, θ_E that minimizes P^N .
- ▶ θ only has four possible values since $\theta(Null) = ND$.
- ▶ θ is thus a function that $\{Pass, Fail\} \mapsto \{D, ND\}$.
- ▶ It's intuitive and can be shown that at equilibrium $\theta(Pass) = D, \theta(Fail) = ND$ as others are dominated strategies.
- ▶ So we only need to search across all x_E assuming $\theta(Pass) = D, \theta(Fail) = ND$.

Third Model

Solution



x: Test Difficulty; left y: y: Novice Belief MSE; right y: P^N

► Similar strategic behavior exhibits for this model.

Epistemic Regulation/Governance

Pros and Cons

- ▶ The model shows that the only plausible remedy for the novice's epistemic performance is imposing a minimal testing standard.
 - ▶ Regulations of this kind are in place in e.g., medical licensing.
- ▶ Potential Concerns: Diversity.
 - ▶ Can come from other sources (Solomon, 2006).

Epistemic Regulation/Governance

Why Regulate?

- ▶ Regulation seems appealing from a pure epistemic standpoint.
- ▶ Ethical considerations?
- ▶ Epistemic virtues at an institutional level

Selected Bibliography

Expert Identification

Collins, H., Evans, R. (2007). *Rethinking expertise*. Chicago: University of Chicago Press.

DeMarzo, P. M., I. Kremer, and A. Skrzypacz (2019). "Test design and minimum standards." *American Economic Review* 109(6), 2173-2207.

Goldman, A. I. (2001). "Experts: Which ones should you trust?". *Philosophy and phenomenological research* 63(1), 85-110.

Hardwig, J. (1985). "Epistemic dependence." *The Journal of philosophy* 82(7), 335-349.

Kamenica, E., and M. Gentzkow (2011). "Bayesian persuasion." *American Economic Review* 101(6), 2590-2615.

Selected Bibliography

Social Epistemology in General

Gilbert, Margaret (1987). "Modelling Collective Belief", *Synthese*, 73(1):185-204.

Gilbert, Margaret (1989). *On Social Facts*, New York: Routledge.

Gilbert, Margaret (2004). "Collective Epistemology", *Episteme*, 1(2):95-107.

Goldman, Alvin I. (1999). *Knowledge in a Social World*. New York: Oxford University Press.

Goldman, Alvin I. (2011). "A guide to social epistemology". In *Social Epistemology: Essential Readings*. Oxford University Press, New York.

Kitcher, Philip (1993). *The Advancement of Science*. New York: Oxford University Press.

Thanks!

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